

Drowned Duct Rescue

Home Energy introduces, with the millennium issue, our new House Doctor column. Send your home performance contracting questions to housedoc@homeenergy.org, and we'll consult experts like these to advise you.

Dear House Doctor:

I would be grateful for any help you can give me in identifying the cause of a wet metal supply duct in a Florida crawlspace. I've suggested some possible causes, with inspection information for each, and although I give my own suggestion below, I'd like your input as well.

Description

The duct segment in question is a round metal supply branch, located in the crawlspace, of a duct system that is otherwise located in the attic and whose trunks are made of duct board. In the attic, a horizontal air handler leads to supply trunks. From the main supply trunk, the third downstream branch makes a 90° turn and travels down a vertical chase to the crawlspace. Once in the crawlspace, this branch runs horizontally for 30 ft, then turns up another vertical chase to terminate in a supply register, which is about 8 ft above floor level. (See Figure 1.) In the horizontal run in the crawlspace, the insulation of the bottom arc of the duct wrap is saturated with water. A small amount of condensate appears on the outside bottom surface, but not on the top surface. The dripping water creates a line of moist soil beneath the full length of this 30-ft branch. A few colonies of tiny frogs have developed along this line of wet soil. Outdoor temperature is about 85°F.

The crawlspace is quite dry (powdery sand, no wood deterioration) and has few vents and no ground cover. The site is well shaded with trees, on a golf course, and the lawn is well watered. At one time, a fan was used to power-vent the space, but the problem occurred at that time as well.

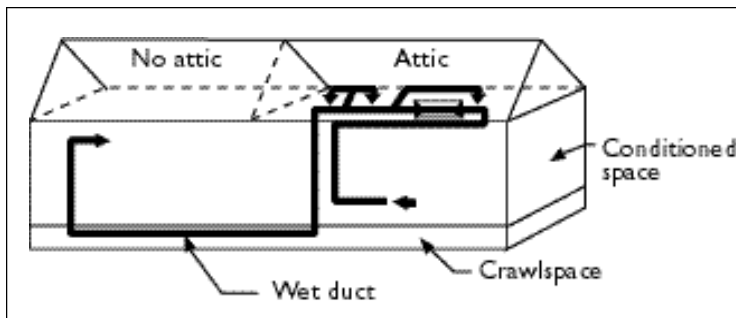


Figure 1. The "drowned" section of the ductwork is located within the crawlspace.

Site Observations

Condensate from the air handler. I cut an opening in the top of the main duct board trunk and determined that no water is traveling from the air handler to the branch in question. There is no evidence (water stains) of past water movement from the air handler to this branch, either from suspended droplets or from overflow flooding.

Unsealed duct. I removed a segment of duct wrap from the crawlspace branch in question and found that the gray duct tape is firmly attached to the joints of the metal air barrier of the duct. So apparently cold supply air alone is not nullifying the insulation properties of the duct wrap, causing condensate to form on the outside of the wrap. Nevertheless, water is between the wrap's vapor barrier and the metal duct.

Uninsulated duct. A look into the vertical chases shows that all vertical and horizontal segments of the branch in question are insulated. Therefore, it doesn't seem that there is condensate on a bare vertical segment that would feed water into the space between the metal duct and the wrap's vapor barrier.

Integral vapor barrier. The foil-covered outer surface of the wrap (which was installed one year ago in an attempt to eliminate the wet duct problem) appears to be continuous, with foil-backed tape closing all joints in the vapor barrier. There is also a second layer of wrap underneath, which was installed 20 years ago, during construction of the house. In the vertical chases, the duct has only one layer of insulation wrap, and the plywood cap at the bottom of one chase is slightly stained with water. The plywood has a round hole for the duct and fits very snug against its wrap, with no missing wrap at this or any other location.

Possible Fix

My suggestion is this: Gaseous water enters the insulation layer by diffusion directly through the foil of the duct wrap, which functions inadequately as the wrap's vapor barrier. These vapor retarders are not impermeable. They have perm ratings of 0.02, and the installation process itself, with much dragging, flexing, and tightening, increases the overall perm rating further. The duct wrap is also imperfectly sealed, allowing moisture to enter at joints and penetrations.

I feel that duct board would be better, since there is no second interior vapor retarder. Any diffuse moisture that passes through the outer FSK foil wo-

Continued on page 8

Continued from page 7

uld enter the airstream and be carried away and removed at the indoor coil. It would not saturate the duct insulation. There would still be condensation on the outside surface, but 1½ inch duct board at R-6 would minimize that. A contributing factor to this outer surface condensation is the low emissivity of the foil, which tends to make the outer surface colder. Black paint may raise the vapor retarder surface temperature, but it would violate the UL listing of the duct product. Further, we might consider the merits of adding weatherproofing atop the duct board's FSK vapor barrier, perhaps EPDM rubber sheeting (joint at bottom), to help protect the foil surface against surface condensate damage.

Philip Wemhoff
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Jacksonville, Florida

The House Doctors Respond:

For this complex problem, three different building scientists have offered several unique solutions.

Leon Neal

My prime suspect for the cause of the saturated insulation around the duct and the dripping from the duct is outside air moving through the insulation around the metal duct, inside the vapor barrier. There are at least two driving forces that could pull outside air into the insulation through any airhole or at the ends of the insulation if the ends are not properly air sealed. The first is buoyant hot air rising toward the high-temperature attic. The second is the possibility that the end of the duct liner is located in a negative-pressure zone. Remember that insulation is not always a barrier to air movement—it is sometimes just a filter.

This drowned duct would not have occurred if the metal duct had been totally sealed with mastic at installation and the vapor wrap of the duct insulation had been totally sealed along its length and at the ends. Regardless of how the air is entering the insulation and what force is moving it through the insulation, the



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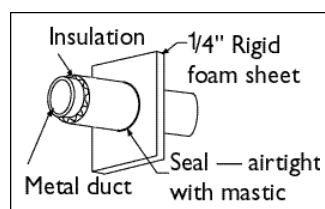


Figure 2. Install rigid foam air barriers in several places along the insulated metal duct to prevent air movement. If no air moves along the ductwork, condensation cannot occur.

cheapest, easiest, simplest solution may be to provide air dams (barriers) at various points along the duct run. This reduces air movement through the insulation along the length of the duct, and reduces the possibility of condensation across the duct surface.

Although I have not tried this concept, this is how it would work: I would probably

place several foam air dams constructed of rigid foam board at easy-to-reach places along this duct run when removing and replacing the saturated current duct liner. I would also seal all duct joints along the metal duct with mastic while I had it bare. Good locations for dams would be at a joint in the attic, at the joint of the elbow in the crawlspace for the down run (one on the vertical and one on the horizontal), and at the joint of the elbow at the start of the up run at the other end of the crawlspace. These five air dams should very much reduce the air moving through the insulation around the metal duct. (See Figure 2.)

The pooling of cold air in this duct when the equipment is not running (because this duct essentially forms a p-trap for the air) may still cause the small amount of condensation on the outside of the vapor barrier, but that does not seem to be a major problem.

Jim Cummings

I can see three possible causes and fixes, but the third one is the one I tend to favor.

Fix 1: During periods when the AC is off, heavier conditioned air may leak from the crawlspace duct, drawing more humid air from the attic (perhaps through the air handler, as these are notoriously leaky)

into the vertical, colder duct sections. In these vertical sections, the humid air may condense on the inside surface of the metal duct. It will drain as liquid water to the lower horizontal crawlspace duct. From there, it leaks out into the insulation jacket. More careful sealing of the crawlspace duct and air handler would tend to disrupt this mechanism.

Fix 2: Every few feet, drill small holes in the metal duct (perhaps at the top). A small amount of supply air would then enter the insulation layer, perhaps exiting at the bottom, where water has created holes in the outer jacket of the duct wrap. The drier supply air would tend to dry out the insulation layer, but unfortunately would tend to increase condensation on the outside of the duct wrap.

Fix 3: Close and seal the crawlspace, and provide about 100 CFM of supply "leakage" to the space (perhaps through an adjustable damper), first at 100 CFM, then, if successful, reduced to 50 CFM. The crawlspace should become semiconditioned, with lower relative humidity and air temperatures.



FSC

John Tooley

Gaseous moisture may move into the insulation of the wrap in either of two ways: (1) directly, through the wrap's vapor retarder; or (2) indirectly, by condensation on the outside of the wrap, which then deteriorates the vapor barrier and enters the insulation as a liquid. The deterioro-



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rated vapor retarder would then allow more gaseous moisture to enter as well, which will condense on the cold metal duct within. The fix would be to seal and condition the crawlspace by means of a continuously operating fan from the house to the crawlspace. This is preferable to an AC supply outlet in the crawlspace, because the semiconditioning of the crawlspace would be independent of the operation of the HVAC system. All crawlspace vents and other openings must be sealed, and a 6-mil vapor retarder must cover the earth and run up the foundation wall. The air supply rate from house to crawlspace would be about 1 CFM/50 ft² of crawlspace floor, about 20 CFM/1000 ft² of floor. Ideally, there should be enough makeup air to the house to replenish the air being exhausted to the crawlspace, but the exhaust amount is so small, in this case, as to be insignificant.

Jim Cummings is a principal research analyst at Florida Solar Energy Center. John Tooley is a building science specialist and Leon Neal is a building science engineer at Advanced Energy in Raleigh, North Carolina.